

the inescapable vehicles of future physical theory, if only as a point of departure. They also are the starting points of attempts to unite quantum theory with Einstein's general theory of relativity, which is a geometrical theory of gravitation as the curvature of space-time. These directions in physics have in turn stimulated further developments in mathematics, notably the theory of "noncommutative geometry." Klein's conjecture continues to be deeply provocative: nonabelian groups may give deep clues to the nature of space and time. As noted in the last chapter, the rotational invariance of three-dimensional space leads to the five Platonic solids. Perhaps such considerations of groups may someday explain why we perceive three dimensions of space and only one of time. The search has only just begun.



Abel's work opened the way to Galois's explanation of the failure to solve all equations in radicals. Indeed, radicals had posed dilemmas since the beginning. The discovery of irrational quantities cast into doubt the nature of mathematics and, with it, the possibility of definitive human knowledge. In Greek, *krisis* means a trial, a decisive act of distinction, separation, judgment. The impossibility of making the irrational commensurate with the rational tested and formed Greek mathematics and philosophy.

Much later, Abel's "impossibility proof" was a watershed for modern mathematics. It had the dimensions of a crisis, though its significance has not been fully acknowledged or assessed. To do so requires a special effort of imagination that would allow us to return to a moment that has been forgotten, the moment in which Abel confronted what he had proved.

As discussed earlier, it is not clear whether the ancients regarded the discovery of the irrational as a disaster or a miracle. It was ominous; one might drown in such an unnameable sea, beyond the security of the counting numbers. As Euclid showed at the end of his Book X, the irrationals

lead to an infinite vista, which could inspire horror or wonder. Euclid tried to keep them rigorously separate from the integers, if only to avoid paradox and confusion. Yet what the ancients separated, the moderns united. Gradually but inexorably, the development of algebra led to the term "number" being applied to irrationals as well as integers, for $x = \sqrt{2}$ seems no less valid a solution of the equation $x^2 = 2$ than $y = 2$ is a solution of $y^2 = 4$.

Once symbolic algebra was established, the crisis of unsolvability was inevitable. Abel's proof showed that radicals are not sufficient, just as the ancient proof showed that rationals are not enough. Since solutions of all equations are known to exist by the Fundamental Theorem of Algebra, they must be a new kind of magnitude. The name "algebraic numbers" came to signify all the solutions of algebraic equations whose coefficients are integers. As with the irrationals, giving them a name seemed tantamount to dispelling the sense of crisis or paradox. For mathematicians, new solutions of the quintic eclipsed the question of solution in radicals, for they could simply annex the algebraic numbers to those expressible in radicals. To give a name to those algebraic numbers that are *not* expressible in radicals, we might call them "ultraradical numbers."

Now we can complete the odyssey of the quintic. It turns out that their solutions can be identified with the so-called elliptic modular functions (defined as sums of powers times trigonometric functions). These functions are based on the "theta functions" that Abel and Jacobi began to study in 1827, though it was not until 1858 that Charles Hermite, Leopold Kronecker, and Fernando Brioschi independently showed their relation to the quintic. By 1870, Jordan had proved that any polynomial equation, of whatever degree, could be solved by using these generalized functions.

During the same period, people started building machines that could find approximate solutions. About 1642, Blaise Pascal had the first vision of machines that could compute. In 1840, Leon Lalanne devised a mechanical computer that could approximate equations up to the seventh degree. In 1895, Leonardo Torres Quevedo developed an elegant machine that could calculate logarithms needed to solve difficult algebraic problems (figure 10.1). At Bell Laboratories in 1937, scientists developed a larger but less elegant-looking machine, the Isograph, which could approximate equations up to the fifteenth degree (figure 10.2). By the end of the twentieth century, far more powerful programs were available to users of personal computers.

These developments simply sidestepped the problem of solution in radicals. But what, then, happened to the “crisis”? Was it ever resolved? Or is it a case of history being written by the victors, the moderns who successfully bypassed and set



Figure 10.1

A machine built in 1895 by Leonardo Torres Quevedo to calculate logarithms.



Figure 10.2

The "Isograph," built in 1938 at Bell Laboratories to solve polynomial equations up to the fifteenth degree.

aside the older scruples? Perhaps they were right not to look back; they were so busy forging ahead that they had little time for retrospection. Yet surely there remains valuable insight to be gained by reflecting on what each successive crisis really meant.

Each crisis opened a new insight into the infinite, in different contexts. First, between arithmetic and geometry: the Greeks learned that expressing an irrational requires an infinite number of digits. Next, between geometry and algebra: to express the area of an oval requires an infinite number of algebraic terms, as Newton showed. Finally, Abel's proof, within algebra itself: solving a finite equation requires an infinite number of terms.

Even solving a cubic equation with real roots requires that we take the cube root of an arbitrary complex number. As

complex numbers were accepted, taking their roots became imperative. The problem was resolved by Abraham De Moivre in 1707 and given its modern form by Euler in 1748. We must start by expressing each complex number in terms of a magnitude and an angle, understood as polar coordinates in a plane. De Moivre's formula then gives a beautiful expression for the n th power of the complex number (where n could be any integer or fraction) in terms of sine and cosine functions (box 10.1).

Most calculators provide sines or cosines by pressing a button. Geometrically, these functions are ratios of the sides and hypotenuse of a right triangle. Yet algebraically sines and cosines are not finite polynomials, but infinite series. Newton had already discerned this algebraic infinitude in studying

Box 10.1

De Moivre's formula

Express the complex number z in terms of its absolute value r and its argument θ , $z = r(\cos \theta + i \sin \theta)$. De Moivre proved that

$$z^n = r^n(\cos n\theta + i \sin n\theta).$$

To show this, consider the two "Taylor series" $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$ and $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$. If we write $z = re^{i\theta}$, where $i^2 = -1$ and $e = 2.718\dots$ is the basis of natural logarithms, then $z^n = r^n e^{in\theta}$. Using the Taylor series for $e^{in\theta} = 1 + in\theta + \frac{i^2 n^2 \theta^2}{2!} + \dots$, De Moivre's formula follows if we group the real and imaginary terms separately:

$$z^n = r^n e^{in\theta} = r^n \left[\left(1 - \frac{n^2 \theta^2}{2!} + \dots \right) + i \left(n\theta - \frac{n^3 \theta^3}{3!} + \dots \right) \right].$$

If we set $n\theta = \pi$ and $r = 1$, then $e^{i\pi} = -1$ (since $\sin \pi = 0$ and $\cos \pi = -1$) or $e^{i\pi} + 1 = 0$, Euler's beautiful formula.

the areas of ovals. For him, this implied that geometry can express in a few lines an infinite series of algebraic terms. Thus, Newton was able to write the sine and cosine as $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$ and $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$. Using such series, calculators sum a few terms to provide a value of required accuracy. As box 10.1 shows, these series lead to the formula $e^{i\pi} + 1 = 0$, one of Euler's most beautiful thoughts, uniting e , i , π , 1, and 0 in one pregnant expression.

Though it was long suspected, finally in 1873 Hermite showed rigorously that e was transcendental, not the solution of any algebraic equation of finite degree, and in 1882 Ferdinand Lindemann did the same for π . These proofs ended the dreams of the circle-squarers, whose hopes rested on the possibility of an algebraic formulation of π . One might have thought that such transcendentals were rare and exotic, but e and π were only the beginning. In 1874, Georg Cantor showed that there are uncountably many transcendentals, which are far more "dense" in the real line than the countable infinitude of integers or of rational numbers. He also showed that algebraic numbers are countable.

To do these later developments justice would require another book and would take us away from Abel, to whom we now return one final time. His discovery showed that there is an intermediate class between algebraic irrationalities (such as the radicals earlier expected to solve the quintic) and the full generality of transcendental numbers. The ultraradical solutions of the quintic equation, in general, are neither the one nor the other; they are algebraic numbers, but they are not composed of radicals. In this, they are a kind of amphibian between land (the radicals) and the vast transcendental sea. Here, they are only one among innumerable kinds of irrational numbers, whose infinitude was already implicit in Euclid's Book X.

Standing on the edge of this ocean, Abel gazed at the dawn of a new mathematics, which he began to explore in the few years remaining him. Great as it was, his proof of the unsolvability of the quintic came during his school years. It would be unjust to represent it as his only or greatest accomplishment in mathematics. He went on to do profound work on transcendental functions, infinite series, and integrals that generalize trigonometric functions, the "Abelian integrals" and "Abelian Addition Theorem" that deeply influenced those who came after him and that many mathematicians call his greatest work. He also took Gauss's seminal insights even further than the master. Where the young Gauss had divided a circle into seventeen parts to construct a regular seventeen-sided polygon, Abel was able to make a similar division of a more complex curve, the lemniscate.

Nevertheless, Abel called the solvability of equations "my favorite subject." Though not much given to speculating about the significance of what he had done, in the introduction of his final paper on the solution of equations (written in 1828 but published only posthumously in 1839) Abel articulated a kind of credo: "One must give to a problem a form such that it is always possible to solve it, which one can always do with any problem. Instead of looking for a relation that one does not know exists or not, one must ask if such a relation is really possible." He had shown that solvability in radicals is a condition that need not always exist. By transcending this limitation, Abel solved the unsolvable. As if taking up Viète's bold claim "to leave no problem unsolved," Abel recorded the power of the infinite extension of finite algebra. The end of the old assumption that all equations have a finite solution revealed a new mathematics of infinite series and noncommutativity.

There is another document that records something of Abel's inner feelings about his work and life, but it is cryptic and private. Written in 1826, during his work on the lemniscate, Abel's doodlings in one of his Paris notebooks (figure 10.3) combine mathematics and reverie in so unguarded

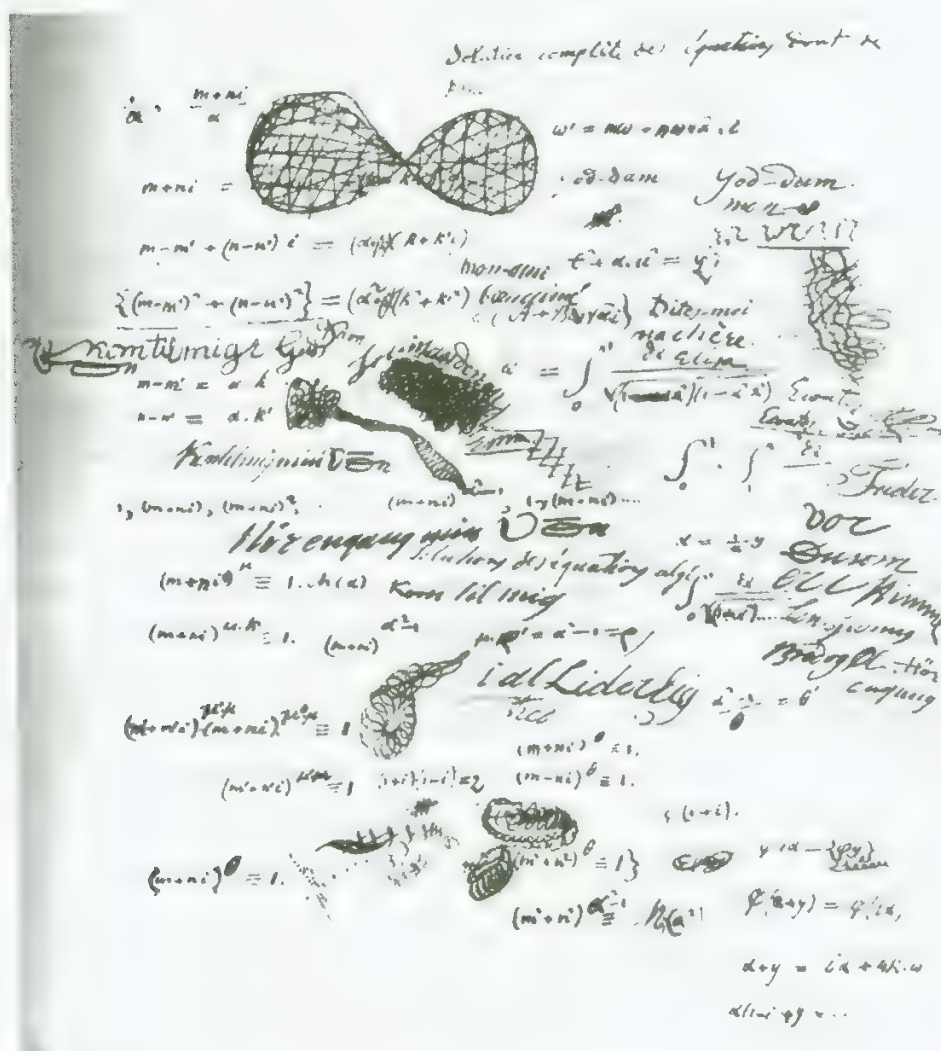


Figure 10.3

A page from Abel's Paris notebook (1826). The large lemniscate (∞) is at the top of the page; the passages discussed in the text are to the right and under it.

a way that he probably would have been amused if not mortified to learn that it would become a treasure of the National Library in Oslo. This page was doubtless meant for no other eyes than his own.

Abel writes in French and Norwegian, interspersed with equations and drawings, featuring a striking drawing of a lemniscate, a large figure ∞ . Abel has cross-hatched this curve and repeatedly outlined it. At the top of this page, Abel writes "complete solution to the equations in which ...," breaking off into equations and intimate thoughts: "My friend ... beloved ... Come to me in God's name" In the midst of this, an ironic prayer: "Our Father who art in Heaven, give me bread and beer. Listen for once." Then, in French, "speak to me, my dear," her name oddly entwined in an elliptic integral,

$$w = \int_0^1 \frac{dx \quad Elisa}{\sqrt{(1-x^2)(1-\alpha^2 x^2)}}.$$

Then: "Listen ... Listen ... Come to me, my friend ... not, for once, my solutions to algebraic equations ... come to me ... in all your lewdness."

No page more daringly suggests the mysterious intersection of desire and mathematics. Who is Elisa? An unknown beloved, one of the tempting Parisians whose beauty he mentioned in a letter, a remembered or even an imagined woman? Why does he scrawl the name of "Soliman den Anden," the Ottoman emperor Sulieman II who also appears on another page? What of the neat equations interwoven among these dreamy fragments? And what of the words near the top of the page: "Goddamn ... Goddamn my ∞ ." What is the tone, the felt meaning of this outburst? Is it humor, vexation, wonder?

We will never know. What remains is equations and ∞ .



Appendix A

Abel's 1824 Paper

Memoir on algebraic equations, in which is demonstrated the impossibility of solving the general equation of the fifth degree (1824)

Geometers have been much concerned with the general solution of equations of the fifth degree and many of them have sought to prove their impossibility, but (if I am not mistaken) none has succeeded until now. Therefore I hope that geometers will receive kindly this memoir, which aims to fill this gap in the theory of algebraic equations.

Let

$$y^5 - ay^4 + by^3 - cy^2 + dy - e = 0 \quad [A1]$$

be the general equation of the fifth degree and let us suppose that it is solvable algebraically, that is, one can express y by a function formed by radicals of the quantities a, b, c, d , and e .

[The commentary is italics in brackets []; equation numbers have been added also in brackets. Note Abel's notation for the coefficients, which does not follow the conventions of our text. He has chosen the signs of [A1] so that the coefficient a is the sum of the roots, b the sum of their products, c the sum of their triples, etc., according to Girard's identities (box 4.1).]

It is clear that in this case we can express y in the form

$$y = p + p_1 R^{\frac{1}{m}} + p_2 R^{\frac{2}{m}} + \cdots + p_{m-1} R^{\frac{m-1}{m}}, \quad [\text{A2}]$$

m being a prime number and R, p, p_1, p_2 , etc. functions of the same form as y , and so on until we come to rational functions of the quantities a, b, c, d , and e .

[For Abel's argument for the generality of [A2], which he considered Step I of the larger proof, see appendix B. Essentially, he shows that a finite sum of radicals, however they are nested, can always be expressed in the form [A2], including putting all the terms on a common denominator, so that eventually one reaches only rational functions inside the innermost radicals. Abel then uses [A2] to frame a reductio ad absurdum: he begins by assuming that y can be written as a finite series of algebraic terms, in which $R, p, p_1, p_2, \dots$ are each algebraic functions of the coefficients a, b, c, d, e , understood in terms of the successive orders of functions as nested radicals, explained in appendix B.]

We can also assume that it is impossible to express $R^{\frac{1}{m}}$ by a rational function of the quantities a, b , etc. p, p_1, p_2 , and by putting $\frac{R}{p_1^m}$ in place of R , it is clear that we can make $p_1 = 1$. Then

$$y = p + R^{\frac{1}{m}} + p_2 R^{\frac{2}{m}} + \cdots + p_{m-1} R^{\frac{m-1}{m}}. \quad [\text{A3}]$$

[Abel simplifies [A2] by getting rid of p_1 : he redefines $R \rightarrow \frac{R}{p_1^m}$ so that $p_1 R^{\frac{1}{m}} \rightarrow p_1 \left(\frac{R}{p_1^m} \right)^{\frac{1}{m}} = R^{\frac{1}{m}}$. In what follows, Abel assumes that this has been done to remove p_1 .]

Substituting this value for y in the equation proposed [A1] and reducing, we obtain a result of this form:

$$P = q + q_1 R^{\frac{1}{m}} + q_2 R^{\frac{2}{m}} + \cdots + q_{m-1} R^{\frac{m-1}{m}}, \quad [\text{A4}]$$

q, q_1, q_2 , etc. being rational and entire functions [i.e., polynomials] of the quantities $a, b, c, d, e, p, p_2, \dots$ and R .

[Now Abel substitutes the presumed solution [A3] back into the main equation, [A1]. Doing so leads to [A4], where the new functions $q, q_1, q_2, \dots$ depend on all the previous quantities: the coefficients a, b, c, d, e and the quantities he has just used, $p, p_1, p_2, \dots, R$. Note that since [A1] sets a polynomial to zero, this rewritten equation also does so, $P = 0$. Note also that, for convenience, Abel has defined $q, q_1, q_2, \dots$ so that each multiplies the appropriate power of $R^{\frac{1}{m}}$. Note also that the highest power of R in [A4] will be $\frac{m-1}{m}$, as shown in appendix B, pp. 173–174.]

In order for this equation to be valid, it is necessary that $q = 0, q_1 = 0, q_2 = 0, \dots, q_{m-1} = 0$.

In fact, calling $R^{\frac{1}{m}} = z$, we have two equations

$$z^m - R = 0 \quad \text{and} \quad q + q_1 z + \dots + q_{m-1} z^{m-1} = 0. \quad [\text{A5}]$$

[Abel sets out to prove $q = 0, q_1 = 0, q_2 = 0, \dots, q_{m-1} = 0$; his proof requires several steps, ending with [A8]. He begins by defining $z = R^{\frac{1}{m}}$, so that $z^m = R$ or $z^m - R = 0$. Also, in [A4] substitute $z = R^{\frac{1}{m}}$, giving $q + q_1 z + \dots + q_{m-1} z^{m-1} = 0$. These two parts of [A5] constrain z .]

If now the quantities $q, q_1, \dots$ are not equal to zero, these equations [A5] necessarily have one or more common roots. If k is the number of these [common] roots, we know that we can find an equation of degree k that has as roots the k roots mentioned and in which all the coefficients are rational functions of R, q, q_1 , and q_{m-1} . Let

$$r + r_1 z + r_2 z^2 + \dots + r_k z^k = 0 \quad [\text{A6}]$$

be that equation. It has these roots in common with the equation $z^m - R = 0$; thus, all the roots of this equation have the form $\alpha_\mu z$, where α_μ designates one of the roots of the equation $\alpha_\mu^m - 1 = 0$. Then substituting [into [A6] $z \rightarrow \alpha_\mu z$], we have